

From cultural codes to spatial measures: A methodological framework for informal settlement analysis

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journals.sagepub.com/home/epb**A. Yair Grinberger¹**  **and Yaara Rosner-Manor^{2,3}** 

Abstract

Informal settlements may appear chaotic, yet often follow internal orders shaped by cultural principles. However, systematic approaches to analyzing and understanding cultural-spatial codes remain lacking. This paper discusses a methodological framework for quantitatively investigating socio-spatial patterns within informal settlements and presents its implementation for Bedouin informal settlements in the Israeli Negev. The framework draws on qualitative data and morphological analysis to examine these settlements' spatio-cultural order. It generates individual and unified measures of adherence to cultural codes, thereby revealing cultural variation across communities. Our implementation demonstrates how Bedouin communities adhere to the Bedouin Spatial Code, revealing structured spatial patterns that persist despite forced relocation and other external pressures. Distinguishing code components by their relationship to the landscape shows that those less tied to physical geography exhibit greater variation, making them more adaptable for planning interventions. This demonstrates how the framework advances the study of cultural-spatial codes, opening avenues for comparative and longitudinal analyses of socio-spatial order in informal settlements.

Keywords

Bedouin settlements, cultural-spatial codes, spatial analysis, settlement morphology

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Data Availability Statement included at the end of the article.

Introduction

Informal¹ settlements occur at multiple scales, from minor modifications to public spaces (Yiftachel, 2009) to large-scale land-use patterns that deviate from formal planning regulations (Dovey et al., 2011, 2018). While often perceived as chaotic, these urban structures frequently emerge from complex, code-based internal orders that may not be immediately apparent (Alexander, 1979; Hakim, 1986, 1994; Nguyen, 2015; Salingaros et al., 2006). These codes represent embedded common knowledge, customs, social structures, and power dynamics. Moreover, the absence of formal constraints may enhance inclusiveness and resilience, allowing communities to adapt their spatio-cultural patterns (Salingaros et al., 2006), with cultural change reflected in spatial transformations and code deviations. While such internal orders are well-documented, variability in code adherence remains underexplored. Understanding such variability is essential for locally sensitive planning, as it allows moving beyond one-size-fits-all solutions toward interventions attuned to internal social and spatial diversity.

Because cultural codes shape settlement layouts (Hakim, 1986, 1994; Nguyen, 2015; Totry-Fakhoury and Alfasi, 2017), they generate distinctive spatial signatures that can be systematically analyzed to explore patterns of variation. Urban morphology offers tools for quantitatively examining spatial form (Dibble et al., 2019) through a range of morphological measures (Dempsey et al., 2010). Previous research has linked spatial form to factors such as ethnic identity (Jia et al., 2020; Nie et al., 2022) and examined the morphological order of informal spaces (e.g., Jones, 2019; Kamalipour, 2016, 2023). Yet, it remains debated whether physical form alone can fully capture underlying cultural logics (Kristjánssdóttir, 2019; Kropf and Malfroy, 2013). However, while such studies show that cultural principles are embodied in spatial form, they do not offer methods for assessing degrees of code adherence or comparing variation across communities. Accordingly, morphometric analyses of cultural adherence require treating cultural codes as an explicit interpretive framework reintroducing the human dimension into morphological analysis (Davis, 2006; Talen, 2009). Doing so is not straightforward, as it requires translating qualitative cultural concepts into measurable terms that fit the logics of morphological analysis.

It is to this gap in identifying and measuring cultural variability in the spatial signatures of informal settlements that this paper contributes. We propose a methodological framework for computing with spatio-cultural codes, which takes qualitatively articulated cultural principles as explicit input and translates them into measurable spatial relations derived from geodata. To enable such measurement, the framework draws on common GIScience stages of spatial representation, operationalized here as four successive stages—conceptualization, formalization, operationalization, and implementation—that progressively reduce ambiguity in moving from cultural meaning to comparable quantitative indicators. The resulting measures allow analysis of spatial variation in code adherence and integration of multiple code components. We demonstrate the approach through a case study of informal Bedouin settlements in the Israeli Negev, examining the Bedouin Spatial Code (BSC).

We proceed as follows: the next section introduces the proposed framework. This is followed by a presentation of the case study and the measurement framework developed for it. The subsequent section presents the results, and the final section concludes with key insights and contributions.

A framework for computing with spatio-cultural codes

Cultural codes, rooted in spatial perceptions and communal knowledge, define spatial forms sustaining specific cultural practices. This embodiment of cultural codes in spatial form is widely recognized. However, existing approaches are not designed to measure degrees of adherence to codes or compare variability and internal differentiation across communities. Visual analysis may reveal whether a code is present, but it does not enable systematic assessment of degrees of adherence across communities. Furthermore, since settlements embody multiple codes with varying

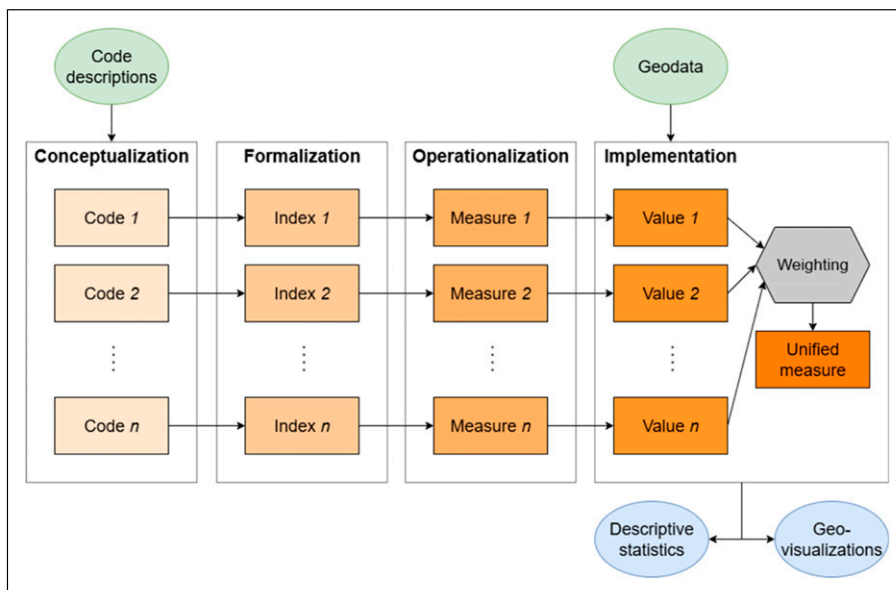


Figure 1. Methodological framework.

degrees of adherence, comprehensively assessing cultural variation through visual methods alone is challenging. These limitations motivate the need for an explicit measurement framework.

The solution proposed here lies in the morphological analysis of geospatial data, which provides quantitative measures for systematic comparison. This enables using weighting methods to assess the relative significance of different codes, ultimately producing a unified similarity measure. However, cultural concepts are inherently qualitative, morphological analysis is quantitative, requiring translation between the two. Moreover, since settlements are physical realizations of ideal cultural codes, resulting measures should capture morphological similarity rather than merely describe form.

GIScience has a rich tradition of converting abstract concepts into computational representations (Schuurman, 2006). We structure our methodological framework to spatio-cultural codes in accordance with this formalization process, dividing it into four stages (Figure 1). Formalization aims to produce computational definitions but is rooted in qualitative foundations—namely, the abstract cultural concepts from which codes are derived. Consequently, our framework is inherently mixed-methods, combining qualitative data on cultural codes and their interpretation (Stage 1) with geospatial data and morphological analysis (Stages 2–4). While the framework is broadly applicable, specific implementations must be tailored to the local cultural codes; thus, the discussion addresses only overarching principles.

The first stage of the framework—conceptualization—uses qualitative inputs to articulate cultural codes in textual form. For example, a code may state that *community boundaries are defined in relation to hydrological units*. This description expresses an inherently spatial relationship, but allows some degree of vagueness—it does not specify the exact form of divisions or the nature of the spatial units.

Formalization translates these qualitative descriptions into explicit spatial relationships, reducing representational uncertainty. In this step, we define *entities* (spatial features such as buildings, roads, or boundaries) and *conditions* (topological or geometric rules) in a structured way, often expressed as an *(entities, condition)* tuple. The condition may specify a relationship between two entity sets, that is *(entities, relation, entities)*. In the hydrological division example, the entities are ownership or community boundaries on one side and drainage basins or streams on the other. The relation is

alignment, meaning community boundaries should follow basin boundaries and/or stream courses, represented as (*community boundaries, alignment, {drainage basins, streams}*).

This description remains normative, indicating what should be—the spatial form fully embodying the code—rather than what exists. Hence, the next stage, operationalization, involves deriving computable morphological measures to quantify adherence to each code. These measures should align directly with the condition/relation component of the tuple. For the boundary alignment code, this may involve calculating alignment indices, such as the Fréchet distance, to measure how closely community borders (implementation) align with watershed boundaries (ideal). At this stage, the geospatial data required for computation is fully defined.

The implementation stage draws on geospatial data to compute individual measures, after which a unified measure is derived through weighted aggregation to assess overall adherence. The final outputs—including adherence maps and statistical measures—offer insight into how cultural codes shape informal settlements. These visualizations uncover spatial logics that might otherwise be overlooked in conventional urban development.

A measurement framework for cultural variation among Bedouin settlements

To demonstrate the application of the framework, we develop a measurement framework tailored to the case of informal Bedouin settlements. Below, we outline their socio-spatial context, then describe the formalization and operationalization of the BSC, the computation of a unified adherence measure, and the case study details required for implementation.

Bedouin “unrecognized” villages: Background

The Bedouin of the Negev are historically semi-nomadic Arab communities with a strong cultural identity shaped by traditional codes, some predating Islamic law (Havakuk, 1986). Over the recent centuries, the Bedouin have undergone a gradual transition to sedentary life (Ben-David, 2004; Yiftachel, 2013). This transition has been influenced by state policies, particularly the displacement of 80%–90% of the population during the 1948 Israeli-Arab war and the subsequent forced relocation of the remaining Bedouin to a restricted area within the Negev (the “Sieg area”). In the 1970-80s, the Israeli government established new settlements for the Bedouin to improve access to education and healthcare while also clearing land for military use, agriculture, and Jewish settlements. However, only about half of the Bedouin population relocated to these towns, with the remainder continuing to live in “unrecognized” villages officially deemed illegal. These villages, which have grown in number and spatial extent, are characterized by harsh living conditions and remain under constant threat of demolition and eviction (Shmueli and Khameisi, 2011; Yiftachel, 2008, 2013).

Numerous studies (Havakuk, 1986; Karplus and Meir, 2013; Marx, 1967; Rosner-Manor et al., 2013; Rosner-Manor and Rofo, 2020) show that Bedouin society is structured around shared rules deeply embedded in community life. Of these, we identify the local cultural principles guiding settlement processes through ideal spatial configurations as the Bedouin Spatial Code (BSC). Processes of modernization and relocation experienced by Bedouin communities may lead to cultural changes, which can manifest as modifications to the code or even its complete abandonment. Conversely, cultural resilience may be reflected in strict adherence to the BSC.

The Bedouin spatial code: Formalization and operationalization

The variety of spatial codes documented in Bedouin settlements (Rosner-Manor et al., 2013, 2020) serves as the qualitative input for our analysis. As a first step in developing a measurement framework

Table 1. Measures of the significance of code elements.

Cultural principle	Concept	Form	Measure	Notation
Territoriality rooted in subsistence and water access	Land is divided based on hydrological units	(Community boundaries, alignment, {basins, streams})	Mean distance of border cells to natural elements	HydroAlign
Climate-risk-aware residential siting	Residence is arranged in relation to streams	(Structures, proximity, streams)	Mean distance of structures to the nearest stream	StreamProx
Collective defense and equal access to climatic resources		(Structures, linear arrangement, streams)	Standard deviation of structure distances from the nearest stream	DwellingLinearity
Kinship-based privacy preservation	Spatial segregation between social units	(Shigs, one-to-one, structure clusters)	Shigs-to-clusters ratio	Separation
Social access regulation	The Shig is the main gateway to a community	(Shig, closest, road)	Mean percentile value of Shig distance from the road, relative to distances of all structures within the Shig's cluster	ShigRoadProx

for the BSC, we selected a manageable subset for analysis (Table 1), based on prevalence in Bedouin settlements and the availability of required geospatial data (see case study section below).

Land is divided based on hydrological units. In the harsh desert environment, survival depends on access to water. Consequently, drainage basins serve as the fundamental living unit for Bedouin communities, with significant social implications: group strength is tied to the size and resource richness of its basins (Ben-David, 2004; Marx, 1973), often compelling resource-poor groups to rely on complex alliances for survival. We formalize this concept as the alignment of community-agreed boundaries (hereafter “community boundaries”) with drainage basin lines and stream paths. Community boundaries often span multiple hydrological elements, crossing several basins or transitioning from basin boundaries to stream paths. Accordingly, we operationalize this code by calculating the mean distance between community boundaries and relevant natural features. We divide the study area into a grid and compute the Euclidean distance from each grid cell intersecting a community boundary (see Appendix, Figure A1) to the nearest hydrological element. For each spatial unit (defined by community boundaries), similarity is measured using the mean Euclidean distance between community boundaries and drainage basin lines or stream paths (equation (1)):

$$HydroAlign_i = \frac{\sum_{j \in N_i} D_j}{|N_i|} \quad (1)$$

where $HydroAlign_i$ is the similarity value for spatial unit i , N_i denotes the set of cells intersecting the boundary of i , and d_j is the Euclidean distance between cell j and the nearest vertex of a hydrological element.

Residence is arranged in relation to streams. Within drainage basins, Bedouin encampments were traditionally located along streams, as foothills provide protection from floods and winter winds (Havakuk, 1986; Marx, 1977). Accordingly, this spatial code is formalized as the placement of structures in proximity to streams, measured by the mean distance of structures from the nearest stream within each spatial unit (equation (2)), with higher values indicating lower adherence.

A related code concerns the linear arrangement of structures, ensuring access to summer winds for ventilation, winter sunlight for heating, and serving defensive purposes². The resulting form consists of structures aligned sequentially along a stream. Accordingly, this code is formalized as the linearity of structure arrangement relative to streams. We operationalize it using the standard deviation of structure distances from streams (equation (3)) since linear arrangements require structures to be located at similar distances from the stream (see Appendix, Figure A2). As before, higher values indicate greater variance, implying weaker adherence.

$$StreamProx_i = \frac{\sum_{j \in B_i} d_j}{|B_i|} \quad (2)$$

$$DwellingLinearity_i = \sigma_{d_j \forall j \in B_i} \quad (3)$$

where $StreamProx_i$ represents the mean distance of structures to streams in spatial unit i , $DwellingLinearity_i$ denotes the standard deviation of these distances, B_i is the set of structures within unit i , and d_j is the distance of structure j from the nearest stream.

The “Shig”—the gathering tent—is the main gateway to the community. In the resource-scarce desert environment, managing tensions and fostering mutual reliance on outsiders are crucial, while controlling the movement of strangers is essential for maintaining household privacy. The Shig plays a key socio-spatial role in this balance, serving as the setting for the coffee ceremony (Havakuk, 1986; Marx, 1977)—a cornerstone of Bedouin social interaction—and as a space for mutual decision-making, hosting, and negotiating power dynamics. The code prescribes positioning the Shig to invite strangers in while restricting access to private residential spaces. Ideally, it is located near a local main road, often marking group boundaries, and on elevated ground, allowing control over traffic while keeping semi-private spaces unobservable and inaccessible.

Formally, this implies that the Shig should be the closest—or among the closest—structures to the main access road within a community. Measuring adherence therefore requires comparing the Shig’s distance to the road relative to that of other structures. We therefore compute the percentile rank of the Shig’s distance relative to all structures within the same community cluster (see Appendix, Figure A3; cluster identification is described in the case study section). As spatial units may contain multiple clusters, we compute the mean percentile value at the unit level (equation (4)). Lower percentile values indicate that Shigs are, on average, among the closest structures to access roads, signifying stronger adherence to the code.

$$ShigRoadProx_i = \frac{P_{j, D_j} \forall j \in S_i}{|S_i|} \quad (4)$$

where $ShigRoadProx_i$ is the mean percentile value for Shig distances to the main access road in spatial unit i , P_{j, D_j} is the percentile rank of the distance of Shig j within D_j , the set of distances of all structures in the community cluster of Shig j , and S_i denotes the set of Shigs within spatial unit i .

Separation between social units. In Bedouin society, geography reflects social divisions, as groups residing along the tributaries of the same stream often share a common origin through patrilineal

kinship within the extended family or the larger clan (“hamis”; Baily, 2000; Yiftachel, 2013). Emphasis on family privacy also shapes this pattern (Hakim, 1986, 1994), fostering clear spatial separation between sub-groups, sometimes even within households.

We formalize this separation as the spatial relationship between structure clusters and Shigs. Since each extended family should have its own Shig, each Shig should ideally correspond to a single structure cluster. A Shig-to-cluster ratio within a spatial unit (equation (5)) below 1 reflects separation even at the sub-household level, while values above 1 indicate insufficient separation and violation of the code.

$$Separation_i = \frac{|S_i|}{|C_i|} \quad (5)$$

where $Separation_i$ is the ratio of Shigs to clusters within spatial unit i , S_i is the set of Shigs located within i , and C_i is the set of structure clusters identified within i .

A unified measure of code adherence

Each measure described above captures a different dimension of the cultural practice of the BSC. However, additional value can be gained by deriving a single metric of overall code adherence per community by linearly combining component values using weights that reflect relevance to the traditional Bedouin lifestyle. Multiple-criteria decision-making offers several weighting approaches for this purpose. Expert-based weights (e.g., from community representatives or officials) are common but are subjective, require translating qualitative judgments into quantitative values, and become problematic when experts disagree (Diakoulaki et al., 1995; Odu, 2019; Saaty, 2013). Data-driven approaches therefore aim to derive weights more objectively.

Here we employ the Entropy Weighting Method (EWM; Odu, 2019) for generating weights. This method assumes that components with greater dispersion contain more information and should receive higher weights (equations 6–8), aligning with our focus on code components that exhibit greater variability, indicating greater flexibility in population practices.

$$P_{y_i} = \frac{y_i}{\sum_{k=1}^m y_k} \quad (6)$$

$$e_y = -\frac{\sum_{i=1}^m P_{y_i} \ln P_{y_i}}{\ln m} \quad (7)$$

$$w_y = \frac{1 - e_y}{\sum_{j=1}^n (1 - e_j)} \quad (8)$$

where P_{y_i} is the proportion of y_i out of the sum of values of y across all units, y_i is the value of feature y for spatial unit i , m is the number of observations, e_y is the entropy value for feature y , w_y is the weight of feature y , n is the number of features.

As EWM provides weights but no scoring or ranking, we combine it with the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS; Han et al., 2024). TOPSIS involves normalizing weighted values (equation (9)), identifying worst- and best-case solutions for each feature (equations 10 and 11), and computing each observation’s distance from both (equations 12 and 13). The final score (equation (14)) approaches zero as the distance to the negative ideal

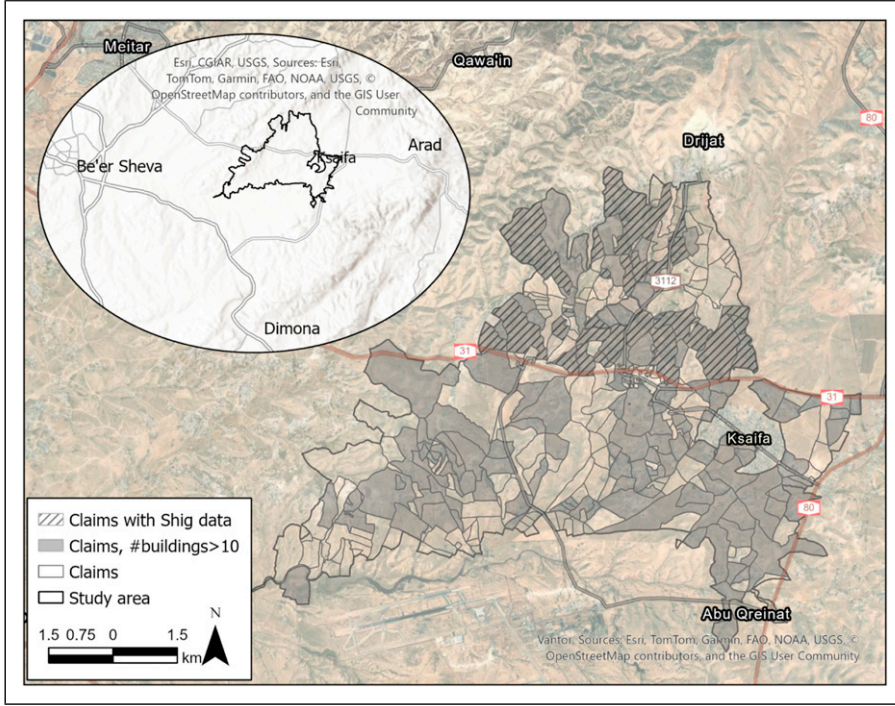


Figure 2. Case study area.

decreases or distance to the positive ideal increases, reflecting weaker adherence. This comparison evaluates observations within empirically observed value ranges and penalizes cases where strong performance on some components masks weak performance on others. We depart from standard TOPSIS by defining the ideal solution with minimum values (equation 10) and the anti-ideal with maximum (equation 11), since lower indicators imply stronger adherence.

$$y'_i = w_y \frac{y_i}{\sqrt{\sum_{k=1}^m y_k^2}} \quad (9)$$

$$V^+ = [y_{1,min}, y_{2,min}, \dots, y_{n,min}] \quad (10)$$

$$V^- = [y_{1,max}, y_{2,max}, \dots, y_{n,max}] \quad (11)$$

$$S_i^+ = \sqrt{\sum_{k=1}^n (y'_{ik} - y_k^+)^2} \quad (12)$$

$$S_i^- = \sqrt{\sum_{k=1}^n (y'_{ik} - y_k^-)^2} \quad (13)$$

$$T_i = \frac{S_i^-}{S_i^- + S_i^+} \quad (14)$$

where y'_i is the normalized and weighted value of feature y for observation i , V^+ and V^- are the best (“positive”) and worst (“negative”) ideal solutions, respectively, S_i^+ and S_i^- are the distances of observation i from the best and worst ideal solution, respectively, y_k^+ and y_k^- are the values for feature y for the best and worst solutions, respectively, T_i is the unified score for observation i .

Case study: Data and specifications

The study area is the Mariit Valley in the Negev Desert (locally known as Makhul *مكحول* and *كحل*), near the planned Bedouin town of Kuseife (Figure 2). This area was chosen because it represents a diverse sample of self-emergent Bedouin settlements. Groups in this region vary in origin, with nomadic heritage alongside other historical trajectories. All settlements developed without formal planning, in proximity to a national road and under external influences, making this a multifaceted and realistic case study.

Implementing the framework requires several geospatial data layers and preprocessing steps (see Appendix, Table A1):

- Community boundaries: We use the “land claims” layer provided by the Authority for Development and Settlement of the Bedouin in the Negev (ADSBN), the governmental body overseeing land and housing issues in Bedouin communities. This layer, originally generated in the 1970s through a state-led survey, is periodically updated from local reports despite lacking regulatory status. Because its polygons were manually drawn at a coarser scale than other layers, we applied a visual correction procedure comparing claims boundaries to basin lines, streams, and roads. Where a boundary closely matched one of these elements in shape but was misaligned, it was shifted to align with the reference element. To limit potential impacts on the analysis, boundaries were adjusted only in cases of clear similarity and author consensus.
- Drainage basin lines and streams: These layers were generated by GeoTeva, an Israeli environmental consultancy firm (<https://www.geoteva.co.il/en>), using spatial and hydrological analyses (e.g., flow accumulation and watershed analysis) applied to a high-resolution digital elevation model (DEM). The parameters were set to detect small sub-basins, reflecting the continuous subdivision of land by Bedouin groups over generations.
- Built structures: We use a national buildings layer compiled by the Survey of Israel and updated in 2022. This dataset is comprehensive, including both formal and informal structures, as confirmed through visual comparison with aerial imagery. For analysis, building polygons were converted to centroids.
- Shigs and access roads: Since no formal data exist, this dataset was generated by a research assistant from the local Bedouin community through fieldwork and resident interactions. Due to time and resource constraints, mapping focused on the inhabited units north of Road 31 (Figure 2). For each mapped unit, the process was exhaustive, yielding a complete dataset.

To ensure consistency in implementation, several analysis parameters were defined:

- Claim filtering: Claims with fewer than 10 structures were excluded.
- Grid cell size in border alignment analysis: A 10×10 m resolution was used to ensure a detailed analysis.
- Clustering of structures: To capture meaningful patterns, clustering must account for environmental sensitivities (e.g., basin size, proximity to other groups) and local variation. We therefore use Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN; McInnes et al., 2017), which accommodates variable density thresholds.

Table 2. Descriptive statistics for the computed measures.

Feature	Mean (std.)	Median (interquartile range)
HydroAlign	17.64 (9.79)	15.53 (10.86)
StreamProx	64.47 (31.02)	56.26 (29.00)
DwellingLinearity	29.74 (13.93)	25.72 (13.53)
ShigRoadProx	0.13 (0.10)	0.11 (0.11)
Separation	0.75 (0.43)	0.57 (0.50)

Clusters were identified using x, y coordinates with Euclidean distance as the metric. The minimum cluster size (4) is based on ADSBN estimates of the minimum number of structures per household.

- Shig distance percentile analysis: Distances were measured from each Shig to the designated access road. Percentile values were calculated relative to all structures within the same cluster, independent of claim boundaries.

The study area includes 350 spatial units (claims), of which 145 meet the structural density threshold and are included in the analysis (41.43% of units, covering 59.72% of the area). Of these, 25 communities north of Route 31 were selected for Shig mapping (7.43% of units). Units meeting the density condition are evenly distributed across the area (Figure 2), limiting potential spatial bias. Restricting Shig mapping to this subset may affect results but reflects a necessary trade-off between analytical coverage and feasible data collection.

Computations were carried out in Python using packages such as geopandas and scikit-learn. The code is available online (<https://github.com/yair-grinb/bedouin-spatial-code-analysis>). Data are not shared to protect community privacy and due to licensing restrictions.

Results

Individual measures of code adherence

Descriptive statistics for individual measures (Table 2) suggest that the BSC is generally upheld. Most measures exhibit low mean and median values, particularly the Shigs-per-cluster measure, where identified clusters exceed Shigs in more than half of the observations. Mean and standard deviation values for stream distance measures are higher, though this is expected given the flood risk associated with proximity to streams. Some variation is nevertheless observed, with standard deviations reaching at least half the mean for some measures, indicating that values twice the mean are not uncommon. This suggests that communities vary in their degree of adherence.

Most measures follow an exponential or quasi-log-normal distribution (Figure A4, Appendix), indicating that a small subset of communities accounts for most variation. The *Separation* measure appears to be an exception, but values equal to or below one can be conceptually grouped, as they indicate good separation, ultimately recreating the overall pattern. Communities with Shig location data show low *StreamProx* and *DwellingLinearity* values, suggesting a bias toward communities more adherent to these codes.

HydroAlign, *StreamProx*, and *DwellingLinearity* are significantly correlated (Pearson correlation: *HydroAlign*–*StreamProx*: $r(145) = 0.34$, $p < 0.001$; *HydroAlign*–*DwellingLinearity*: $r(145) = 0.24$, $p = 0.003$; *StreamProx*–*DwellingLinearity*: $r(145) = 0.70$, $p < 0.001$; see Appendix, Figure A4), indicating a shared adherence pattern. Interestingly, these measures are directly linked to landscape features, in contrast with the *ShigRoadProx* and *Separation*, which relate to Shigs, structures, and

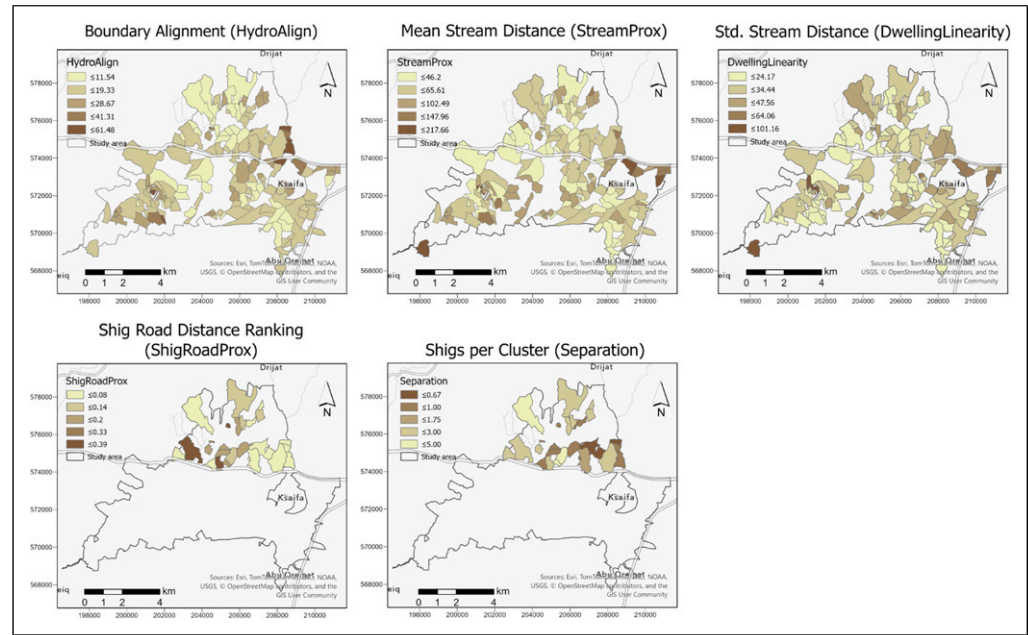


Figure 3. Spatial distribution of measure values. The first three measures exhibit similar spatial patterns, particularly in areas with high values (indicating low adherence), whereas the last two measures display greater spatial variability.

roads—second-order derivatives of these relationships. Notably, landscape-related codes reflect environmentally oriented principles (water-based subsistence, risk awareness, access to climatic resources), while the remaining two serve primarily social functions—separation and regulation.

Spatial patterns reinforce these findings (Figure 3), with certain communities in the southwest of the study area and around Kuseife consistently showing high values across *HydroAlign*, *StreamProx*, and *DwellingLinearity*. Other communities, however, do not exhibit the same consistency across these measures. Additionally, the spatial patterns of the Shig-related measures differ from those observed north of Route 31 and do not align with each other. This highlights variability in BSC adherence across communities and underscores the challenge of assessing code significance within each community, reinforcing the need for a unified measure.

Unified measure of code adherence

We conduct the EWM-TOPSIS analysis in three ways: first, using only measures directly derived from relationships with physical features (*HydroAlign*, *StreamProx*, *DwellingLinearity*—hereafter first-derivative measures), which differ from those considered second-order derivatives of these relationships (*ShigRoadProx*, *Separation*—hereafter second-derivative measures); second, incorporating all measures; and third, applying only first-derivative measures to units north of Road 31 to assess potential selection bias.

When only first-derivative measures are considered, *HydroAlign* receives the highest EWM weight (Table 3), while the strong correlation between *StreamProx* and *DwellingLinearity* reduces their information content and thus their weights, indicating greater variation in subsistence-based territoriality than in climatic considerations. This pattern remains when analyzing only the units north of Route 31, though the weight difference between *StreamProx* and *DwellingLinearity*

Table 3. Entropy-based weights for the code-based measures under three scenarios: (a) first-derivative measures only; (b) first-derivative measures for units north of Route 31; and (c) all measures for units north of Route 31. The strong correlation between *DwellingLinearity* and *StreamProx* results in lower weights for both. When included, social measures (*ShigRoadProx* and *Separation*) receive relatively high weights, with *ShigRoadProx* surpassing *HydroAlign*.

Measure	(a)	(b)	(c)
HydroAlign	0.42	0.59	0.26
StreamProx	0.30	0.23	0.10
DwellingLinearity	0.28	0.18	0.08
ShigRoadProx	--	--	0.32
Separation	--	--	0.23

increases. When all measures are included, *HydroAlign* is surpassed by *ShigRoadProx* and almost matched by *Separation*, while *StreamProx* and *DwellingLinearity* weights remain small and similar. Thus, social considerations appear more variable than physical-environmental ones.

While mean unified scores for units north of Route 31 are lower than those for the full study area (Figure 4(a) and (b)), similar medians and non-normal distributions indicate minimal bias. Medians and distributions also indicate that including second-derivative measures significantly alters adherence levels. Scenarios limited to first-derivative measures exhibit similarly skewed distributions, with most values concentrated near 1 (Figure 4(a) and (b)). In contrast, the scenario incorporating all measures produces a distribution peaking at a more moderate adherence level, with a secondary peak appearing at higher values (Figure 4(c)).

Spatially, when only first-derivative measures are considered, lower adherence values occur in peripheral areas (Figure 4(d)). This does not indicate a clear core-periphery pattern, as high-adherence communities are not clustered, and low-adherence ones are located near top-down governmental developments such as Kuseife, main roads, and the Nevatim airfield. These patterns remain consistent when only units north of Route 31 are considered, with lower adherence communities concentrated in the eastern part of the study area (Figure 4(e)). When all measures are incorporated, some previously high-adherence communities drop in ranking (Figure 4(f)), reflecting the influence of second-derivative measures.

To assess pattern dependence on the weighting scheme, we conducted a sensitivity analysis based on random weight perturbations. In each iteration, two weights were randomly selected and perturbed by adding a random value between -0.1 and 0.1 to one and subtracting the same magnitude from the other. Computing root mean square error (RMSE) values across 10,000 iterations and three scenarios shows minimal changes in the unified score (Appendix, Figure A5), with median RMSE values of 0.008 (first-derivative measures only), 0.010 (first-derivative measures, north of Route 31), and 0.015 (all measures, north of Route 31).

Discussion and conclusions

This paper advances the study of informal settlements by outlining a framework for representing how cultural codes shape spatial organization. Drawing on established GIScience practices, the framework translates qualitative codes into measurable forms through conceptual description, formalization of topological relationships between spatial entities, operationalization of quantitative measures, and implementation using available geodata.

We demonstrate the utility of this approach by developing a measurement framework that translates components of the Bedouin Spatial Code into quantitative indicators. We find that the

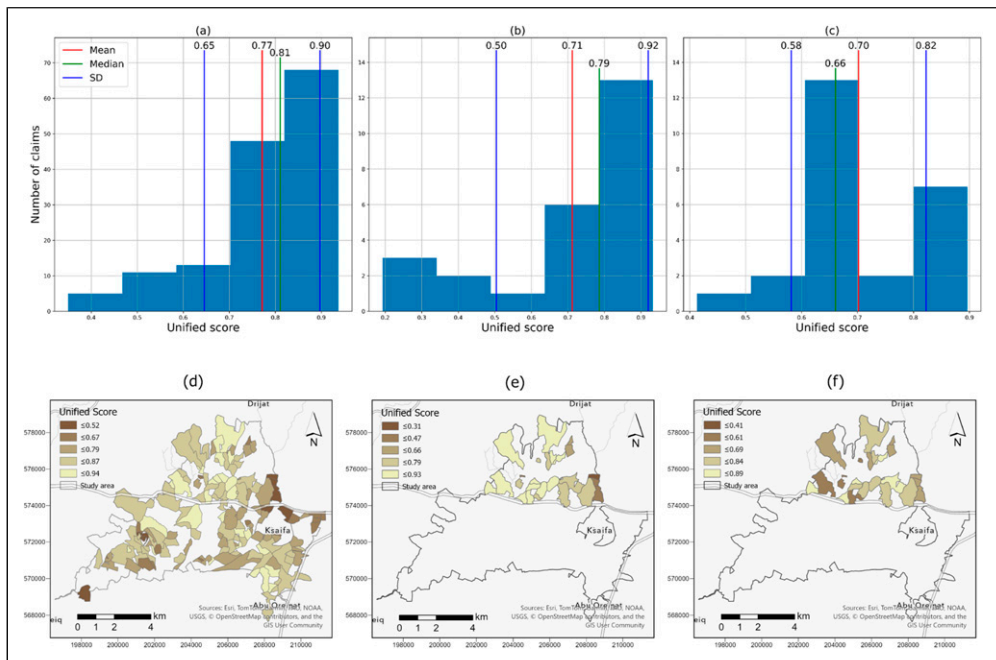


Figure 4. Distributions (a–c) and spatial patterns (d–f) of the unified adherence measure under three scenarios: using only first-derivative measures (a, d); using only first-derivative measures for units north of Route 31 (b, e); and using all measures for units north of Route 31 (c, f). Scenario (c) differs from (a) and (b) by exhibiting a bimodal distribution and a greater concentration of lower adherence values. The similarity between (d) and (e) indicates limited bias from restricting the analysis to units north of Route 31, whereas the inclusion of second-derivative measures substantially alters the spatial pattern in (f).

BSC remains a key organizing principle in Bedouin informal settlements, sustaining cultural identity and institutions despite radical lifestyle changes and repeated forced relocations (broader social and ethical implications of this lie beyond the scope of this analysis). Further insights arise from distinguishing between first-derivative measures, directly tied to landscape features, and second-derivative measures, derived from relationships among built and socio-spatial elements. Deviations from first-derivative codes occur among communities occupying areas reshaped by infrastructure development due to historical and cultural ties to the land. Second-derivative measures, associated with more easily modifiable features, such as the Shig's location, display greater variability.

Based on the premise that spatial patterns reflect meaningful cultural values, the framework helps identify stable and robust codes. These insights—particularly the computed code weights—provide planners with a grounded basis for interventions that may help anticipate and reduce conflicts between state objectives and local practices. Using adherence maps, these insights can be extended to finer resolutions, supporting tailored solutions to groups of communities showing similar patterns.

More broadly, the results illustrate how an appropriate interpretive framework can enrich morphological analysis, yielding proxies for cultural dynamics. The methodological framework thus provides a broadly applicable conceptual roadmap for developing context-specific measurement frameworks for cultural variation through the sequential stages of conceptualization, formalization, operationalization, and implementation, culminating in the integration of multiple codes into a single measure.

We emphasize that the framework's outputs should not be seen as definitive representations of cultural identity. As with any empirical implementation, data limitations and analysis decisions may affect results, including the coarse resolution of community boundary data and its manual correction, the exclusion of small communities, and restricting Shig mapping to part of the study area. These choices were necessary to ensure analytical robustness. Analyses restricted to first-derivative measures north of Route 31 indicate that this subset is broadly representative with respect to first-derivative patterns. More generally, it is worth noting that not all cultural codes have a spatial form, and that our weighting scheme reflects information content rather than community perceptions—a significant choice given that cultural codes are inherently flexible and open to interpretation. This potentially limits how fully cultural diversity as experienced by members of local communities is captured.

The framework's applicability depends on several conditions: (a) the presence of commonly recognized codes with a spatial manifestation—a condition likely widespread, given their role in settlement stability; (b) prior knowledge of these codes and their characteristics—while this might seem to require extensive fieldwork, existing academic studies (e.g., in Bangladesh: [Parvin et al., 2022](#); Papua New Guinea: [Stasch, 2013](#); China: [Jia et al., 2020](#); Vietnam: [Nguyen, 2015](#)) already provide a knowledge basis for applications; (c) availability of suitable geodata—often obtainable from widely accessible datasets, as in our study, thereby reducing the need for dedicated data collection.

Upon implementation, the interpretation of adherence-based outputs requires careful validation to ensure that emerging narratives align with local self-definitions and spatial practices. Importantly, the framework does not define what constitutes “high adherence”; such distinctions lack intrinsic normative meaning and require local interpretation and validation. In this sense, analyses of stability and variation may help articulate patterns and values that communities themselves may find difficult to phrase in formal or spatial terms, but whose legitimacy ultimately depends on local recognition. The framework thus supports deeper participation by local actors, providing a shared vocabulary of values, codes, and spatial forms that communities can use—often with support from local organizations—to formalize their claims and priorities in dialogue with planning authorities. Such participation is essential for responsible use of the framework, as otherwise adherence-based indicators may be (inadvertently) mobilized to justify development trajectories that undermine community welfare or social cohesion.

This emphasis on participation points to a broader research agenda: translating the framework's outputs into inclusive planning strategies and actionable interventions. Alongside this challenge, the framework also enables the study of temporal changes in code adherence and of contexts where diverse populations with distinct codes share space, raising questions of integration, segregation, or the dominance of one code. While rooted in geoinformatics and spatial analysis, outputs of the formalization stage may also inform the development of indigenous geo-ontologies ([Wellen and Sieber, 2013](#)), helping to reveal context-specific patterns.

In a world where informal urbanization is rapidly reshaping landscapes, our framework offers a lens for understanding how cultural logics shape, adapt, and coexist in shared spaces. By grounding computational outputs in cultural meaning, it opens new possibilities for envisioning urban growth that aligns with community welfare while preserving spatial-cultural diversity.

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Ethical considerations

There are no human participants in this article and informed consent is not required.

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Declaration of conflicting interests

The authors declared the following potential conflicts of interest with respect to the research, authorship, and/or publication of this article: YRM is a planner with extensive experience in the planning of Bedouin villages for formal recognition, including some villages within the study area.

Data Availability Statement

The code for the analyses included in the paper is available online (<https://github.com/yair-grinb/bedouin-spatial-code-analysis>). We do not share the data themselves to protect community privacy and due to licensing restrictions.

Declaration of generative AI and AI-assisted technologies in the writing process

ChatGPT was used during manuscript preparation to support language editing. The authors fully reviewed and revised the text and take responsibility for the content.

Notes

1. While the term “informal construction” encompasses diverse categories—including variations in legal status, property ownership, and spatial coherence—we use it here for simplicity and consistency.
2. Historically, tents were sometimes arranged in a square around a shared space for property and livestock, so high *DwellingLinearity* values may still reflect adherence to the code. Given our results, any resulting bias is likely minimal.

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Appendix

From cultural codes to spatial measures: A methodological framework for informal settlement analysis

Table AI. Geodata layers and their sources.

Data	Layer	Source
Community borders	Land claims	ADSBN
Drainage basins	Drainage basin polygons	GeoTeva
Streams	Stream polylines	
Structures	Buildings layer	Survey of Israel
Shigs	Shig locations	Interactions with local residents, field mapping
Access roads	Road polylines	

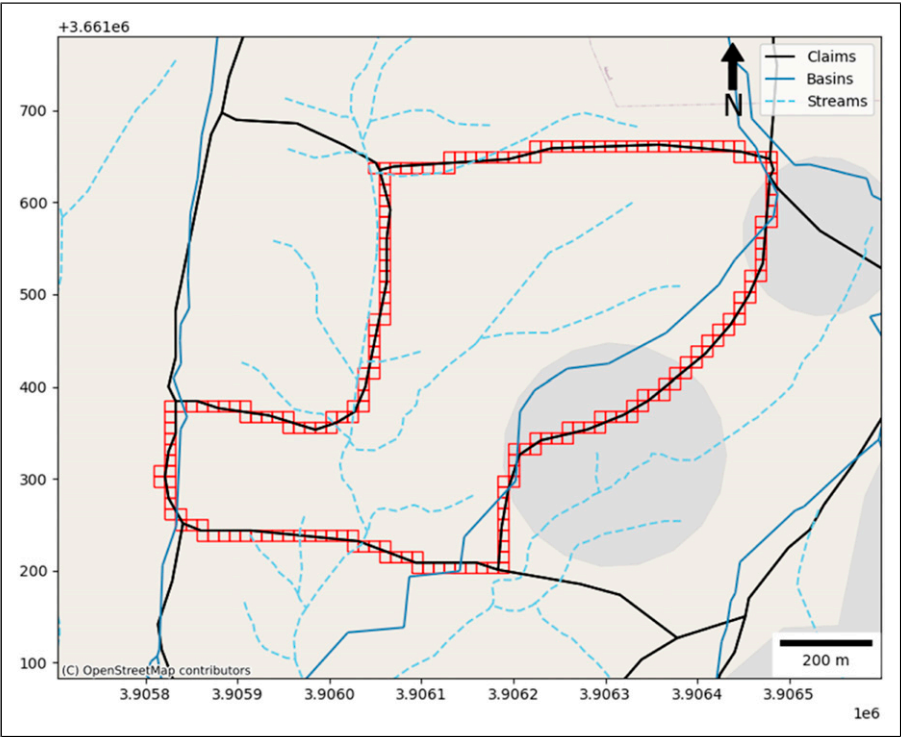


Figure AI. Demonstration of the process for computing the BCD measure.

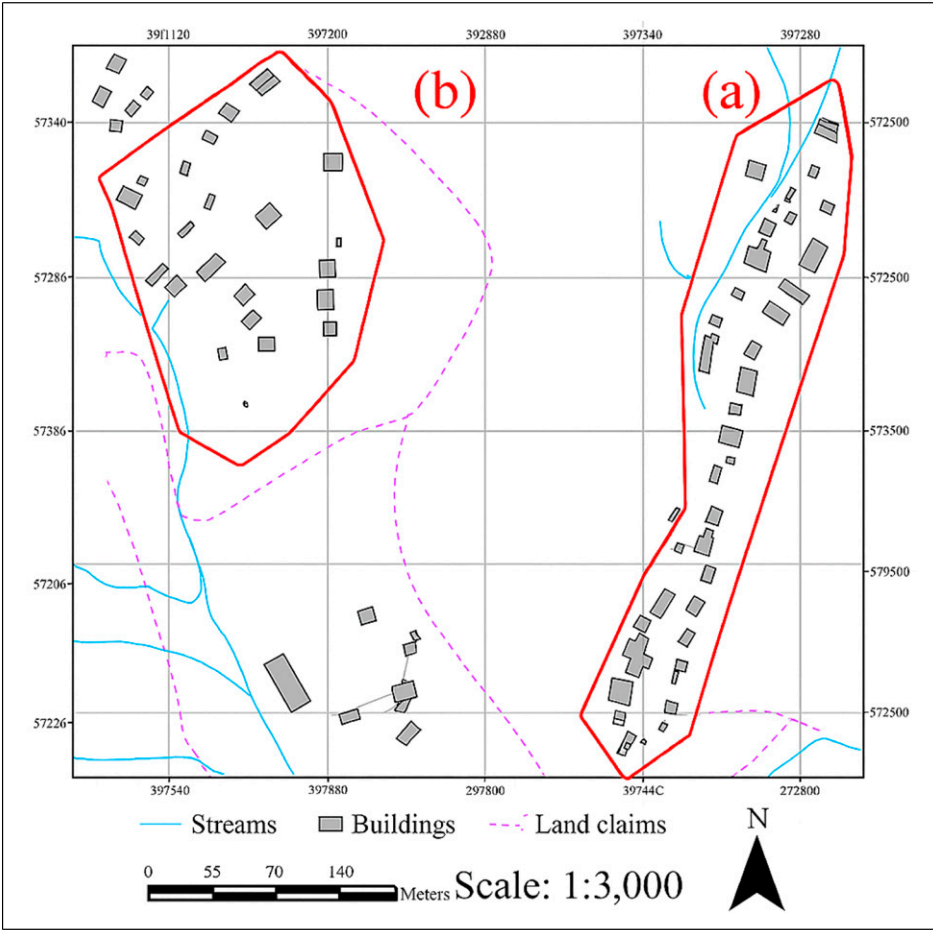


Figure A2. A cluster showing a linear (a) pattern and a non-linear (b) cluster.

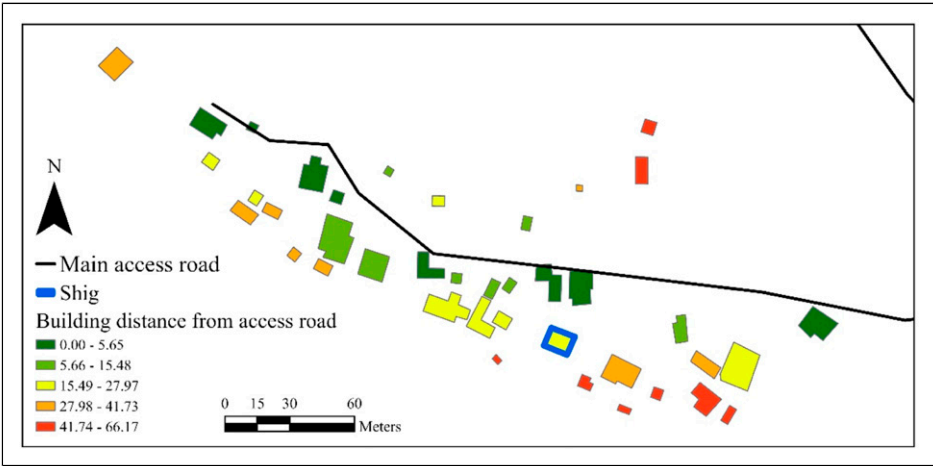


Figure A3. Shig's distance to main access road for a selected community.

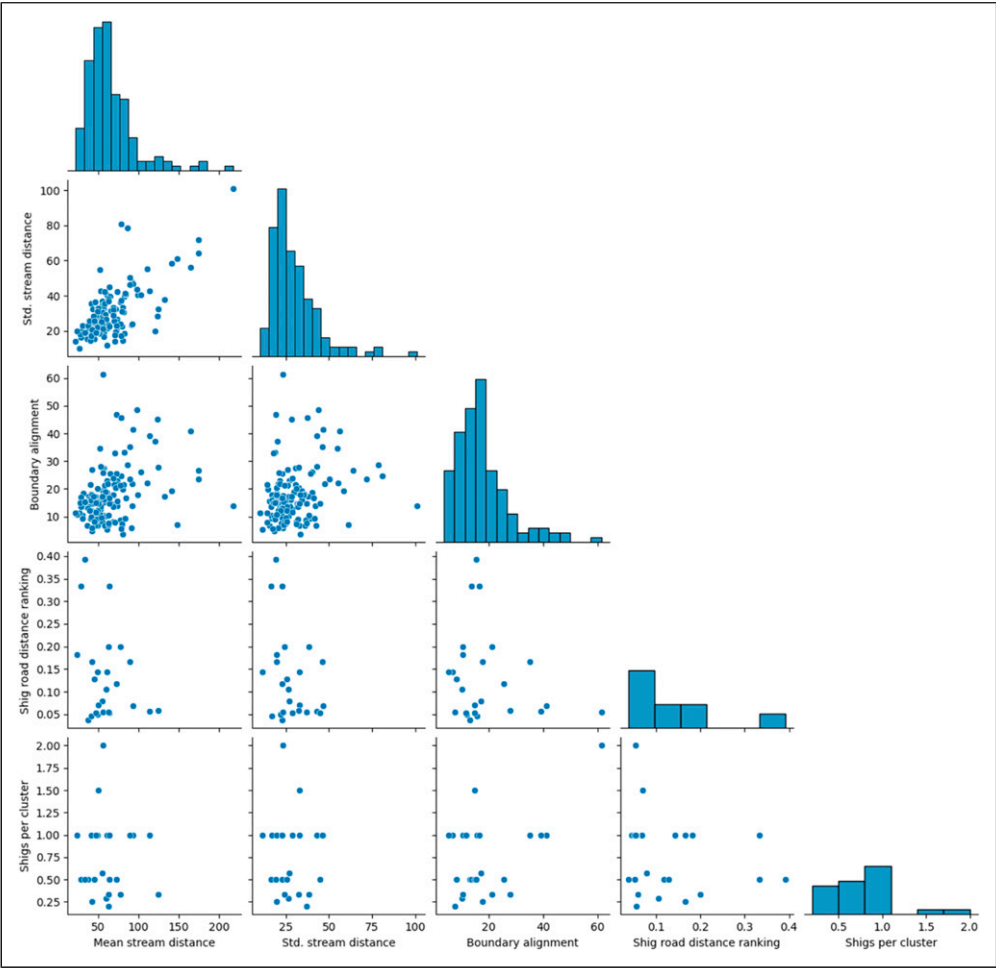


Figure A4. Distributions and joint distributions of code-based measures. For consistency with other measures, where lower values indicate higher adherence, the Shig-per-cluster values are inverted and presented as the number of clusters per Shig.

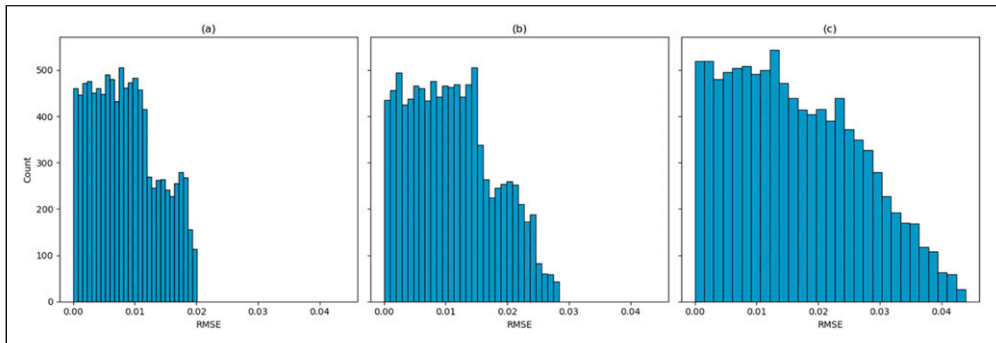


Figure A5. Distributions of RMSE values across 10,000 experiments for three scenarios: (a) first-derivative measures, all communities; (b) first-derivative measures, communities north of Route 31 only; and (c) all measures, communities north of Route 31 only.